

TOPIC: JOINS AND SET OPERATIONS

```
mysql> CREATE TABLE class_info (  
->     id INT,  
->     address VARCHAR(30)  
-> );  
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> INSERT INTO class VALUES  
-> (1,'abhi'),  
-> (2,'adam'),  
-> (4,'alex');  
Query OK, 3 rows affected (0.01 sec)  
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO class_info VALUES  
-> (1,'DELHI'),  
-> (2,'MUMBAI'),  
-> (3,'CHENNAI');  
Query OK, 3 rows affected (0.01 sec)  
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM class;  
+-----+-----+  
| id  | name |  
+-----+-----+  
| 1   | abhi |  
| 2   | adam |  
| 4   | alex |  
+-----+-----+  
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM class_info;  
+-----+-----+  
| id  | address |  
+-----+-----+  
| 1   | DELHI   |  
| 2   | MUMBAI  |  
| 3   | CHENNAI |  
+-----+-----+  
3 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM class
-> CROSS JOIN class_info;
```

id	name	id	address
4	alex	1	DELHI
2	adam	1	DELHI
1	abhi	1	DELHI
4	alex	2	MUMBAI
2	adam	2	MUMBAI
1	abhi	2	MUMBAI
4	alex	3	CHENNAI
2	adam	3	CHENNAI
1	abhi	3	CHENNAI

```
9 rows in set (0.01 sec)
```

```
mysql> INSERT INTO class VALUES
-> (1,'abhi'),
-> (2,'adam'),
-> (4,'alex');
```

Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

```
mysql> SELECT *
-> FROM class
-> INNER JOIN class_info
-> ON class.id = class_info.id;
```

id	name	id	address
1	abhi	1	DELHI
1	abhi	1	DELHI
2	adam	2	MUMBAI
2	adam	2	MUMBAI
1	abhi	1	DELHI
1	abhi	1	DELHI
2	adam	2	MUMBAI
2	adam	2	MUMBAI

```
8 rows in set (0.00 sec)
```

```
mysql> SELECT class.name,
->         class_info.address
-> FROM class
-> INNER JOIN class_info
-> ON class.id=class_info.id;
```

name	address
abhi	DELHI
abhi	DELHI
adam	MUMBAI
adam	MUMBAI
abhi	DELHI
abhi	DELHI
adam	MUMBAI
adam	MUMBAI

8 rows in set (0.00 sec)

```
mysql> INSERT INTO class_info VALUES
-> (7,'NOIDA'),
-> (8,'PANIPAT');
```

Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

```
mysql> SELECT *
-> FROM class
-> LEFT OUTER JOIN class_info
-> ON class.id=class_info.id;
```

id	name	id	address
1	abhi	1	DELHI
1	abhi	1	DELHI
2	adam	2	MUMBAI
2	adam	2	MUMBAI
4	alex	NULL	NULL
1	abhi	1	DELHI
1	abhi	1	DELHI
2	adam	2	MUMBAI
2	adam	2	MUMBAI
4	alex	NULL	NULL

10 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM class
-> LEFT JOIN class_info
-> ON class.id=class_info.id
-> WHERE class_info.id IS NULL;
```

id	name	id	address
4	alex	NULL	NULL
4	alex	NULL	NULL

2 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM class
-> RIGHT OUTER JOIN class_info
-> ON class.id=class_info.id;
```

id	name	id	address
1	abhi	1	DELHI
1	abhi	1	DELHI
2	adam	2	MUMBAI
2	adam	2	MUMBAI
NULL	NULL	3	CHENNAI
1	abhi	1	DELHI
1	abhi	1	DELHI
2	adam	2	MUMBAI
2	adam	2	MUMBAI
NULL	NULL	3	CHENNAI
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

12 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM class
-> RIGHT JOIN class_info
-> ON class.id=class_info.id
-> WHERE class.id IS NULL;
```

id	name	id	address
NULL	NULL	3	CHENNAI
NULL	NULL	3	CHENNAI
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

4 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM class
-> LEFT JOIN class_info
-> ON class.id = class_info.id
->
-> UNION
->
-> SELECT *
-> FROM class
-> RIGHT JOIN class_info
-> ON class.id = class_info.id;
```

id	name	id	address
1	abhi	1	DELHI
2	adam	2	MUMBAI
4	alex	NULL	NULL
NULL	NULL	3	CHENNAI
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

6 rows in set (0.00 sec)

```
mysql>
mysql> CREATE TABLE second_table(
-> id INT,
-> name VARCHAR(30)
-> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql>
mysql> INSERT INTO first_table VALUES
-> (1,'abhi'),
-> (2,'adam');
Query OK, 2 rows affected (0.00 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> INSERT INTO second_table VALUES
-> (2,'adam'),
-> (3,'chester');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM first_table
-> UNION
-> SELECT * FROM second_table;
+-----+-----+
| id    | name    |
+-----+-----+
| 1     | abhi    |
| 2     | adam    |
| 3     | chester |
+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM first_table
-> UNION ALL
-> SELECT * FROM second_table;
+-----+-----+
| id    | name    |
+-----+-----+
| 1     | abhi    |
| 2     | adam    |
| 2     | adam    |
| 3     | chester |
+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql>
mysql> SELECT COUNT(*)
-> FROM
-> (
-> SELECT * FROM first_table
-> UNION ALL
-> SELECT * FROM second_table
-> ) A;
+-----+
| COUNT(*) |
+-----+
| 4        |
+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM first_table
-> INTERSECT
-> SELECT * FROM second_table;
+-----+-----+
| id    | name |
+-----+-----+
|      2 | adam |
+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT name FROM first_table
-> INTERSECT
-> SELECT name FROM second_table;
+-----+
| name |
+-----+
| adam |
+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT f.*
-> FROM first_table f
-> LEFT JOIN second_table s
-> ON f.id = s.id
-> AND f.name = s.name
-> WHERE s.id IS NULL;
+-----+-----+
| id    | name |
+-----+-----+
|      1 | abhi |
+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT name
-> FROM first_table
-> WHERE name NOT IN (
->     SELECT name
->     FROM second_table
-> );
+-----+
| name |
+-----+
| abhi |
+-----+
1 row in set (0.00 sec)
```

```
mysql> SELECT c.id,c.name,ci.address
-> FROM class c
-> INNER JOIN class_info ci
-> ON c.id=ci.id;
```

id	name	address
1	abhi	DELHI
1	abhi	DELHI
2	adam	MUMBAI
2	adam	MUMBAI
1	abhi	DELHI
1	abhi	DELHI
2	adam	MUMBAI
2	adam	MUMBAI

8 rows in set (0.00 sec)

```
mysql> SELECT c.id,
-> c.name,
-> CASE
-> WHEN ci.address IS NULL
-> THEN 'Address Missing'
-> ELSE 'Address Available'
-> END AS Status
-> FROM class c
-> LEFT JOIN class_info ci
-> ON c.id=ci.id;
```

id	name	Status
1	abhi	Address Available
1	abhi	Address Available
2	adam	Address Available
2	adam	Address Available
4	alex	Address Missing
1	abhi	Address Available
1	abhi	Address Available
2	adam	Address Available
2	adam	Address Available
4	alex	Address Missing

10 rows in set (0.00 sec)